

Genetic Non Linear Polynomial Discriminant Analysis GNLPDA

Director: Héctor Gabriel Acosta Mesa
Codirector: Adriana Laura López Lobato

Ángel García Báez
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For this session

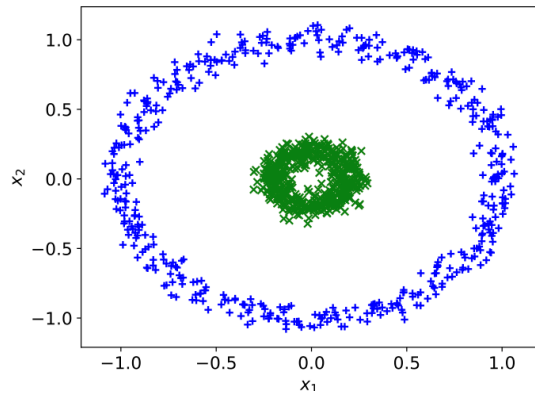
- Show workflow progress
- Show results with statistical test

Calendar January - September 2026									
Activity	2026								
	January	February	March	April	May	June	July	August	September
Review of the protocol									
Review of the literature and state of the art									
Problem definition and methodology									
Design and implementation of the proposed algorithm									
Experiments and validation (using real and simulated data)									
Calibration of the proposed algorithm									
Writing of chapters and figures									
Writing and revision of Chapter 1									
Writing and revision of Chapter 2									
Writing and revision of Chapter 3									
Review by the thesis committee									
Writing an article based on the findings of the thesis									
Thesis defense									
Research internship									

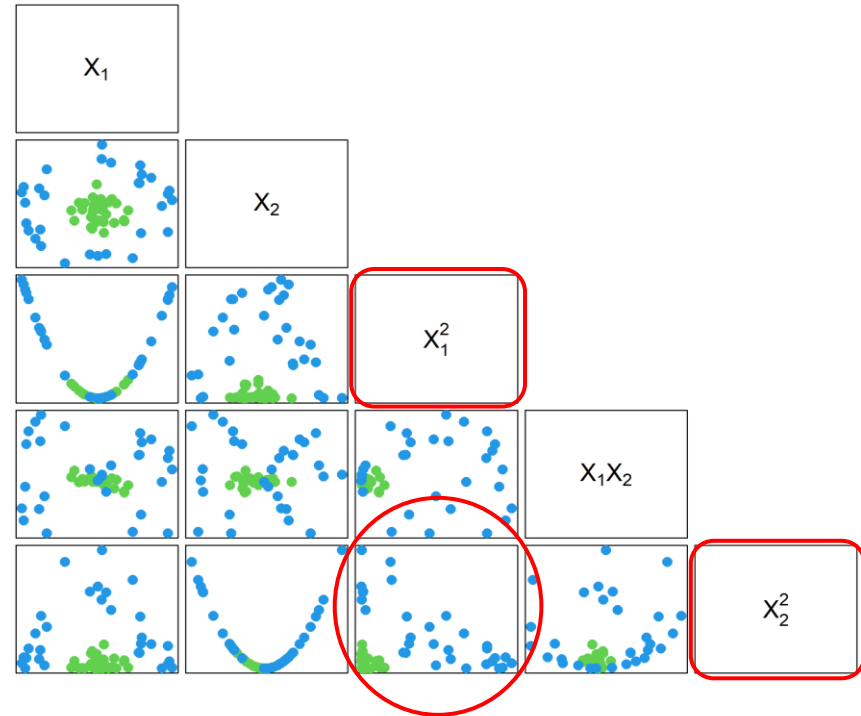
Basic information

- Thesis main goal
 - Develop and implement a feature extraction framework that combines polynomial expansion with classical dimensionality reduction LDA, using an intelligent selection mechanism, to improve the separability and visualization of nonlinear data.
- Thesis hypothesis
 - A polynomial expansion will allow us to achieve levels of accuracy statistically equivalent to those of the most advanced nonlinear models, with the added advantage of filtering out dimensional redundancy to generate visualizable 1D, 2D, and at most 3D projections that reveal the dispersion structure of the classes.

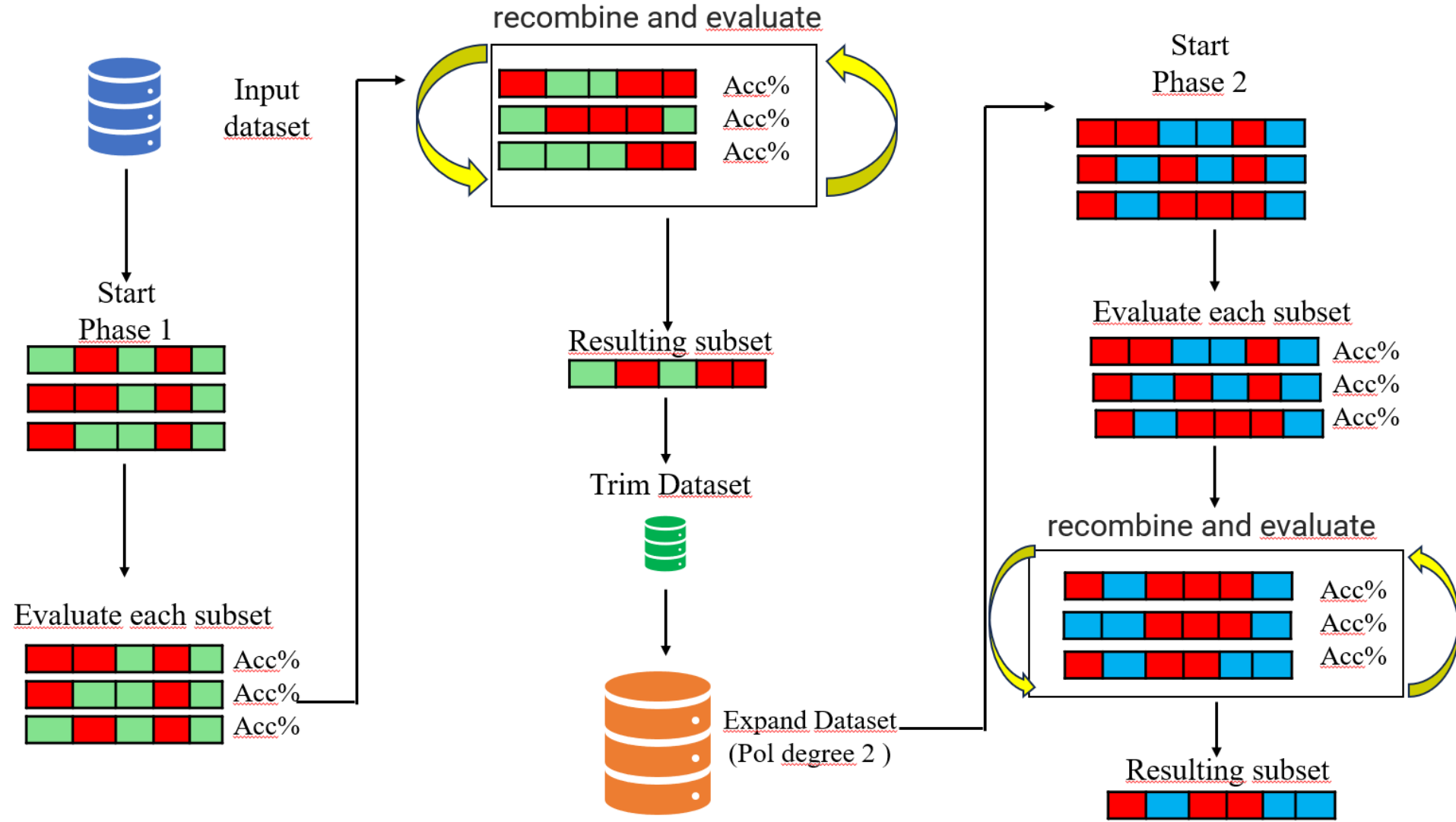
Expectations



A	B	A ²	B ²	AB



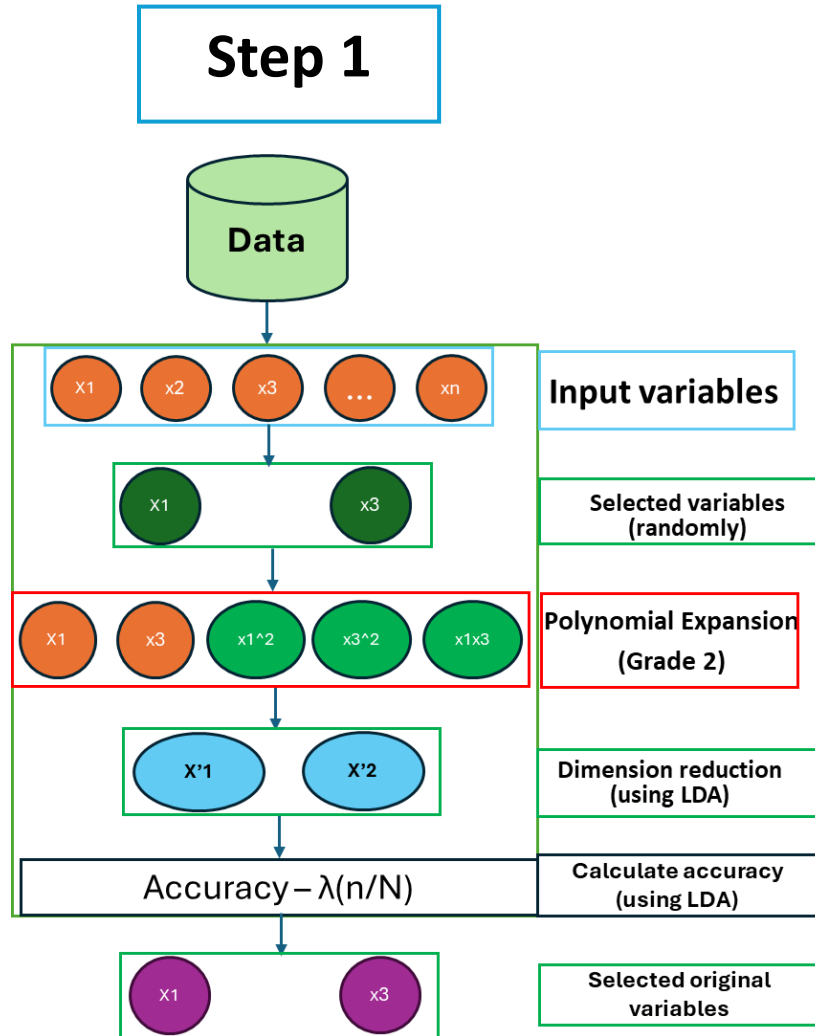
Workflow



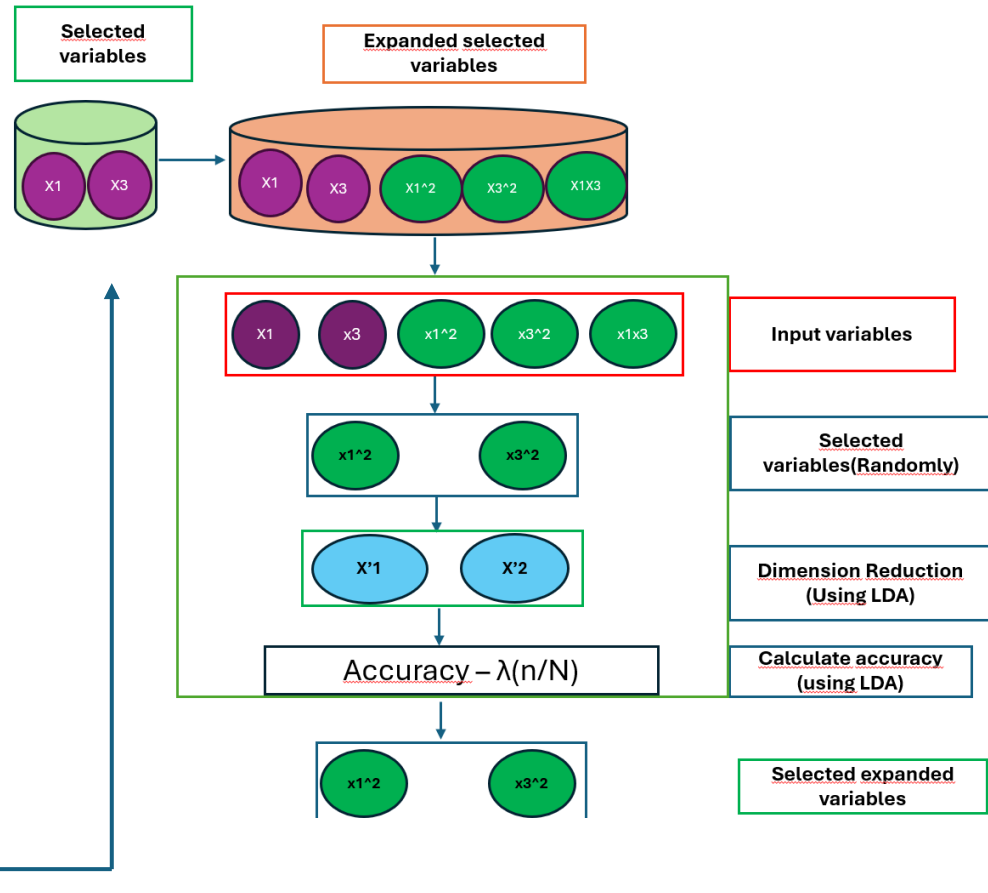
Proposed algorithm

Unified workflow

Step 1

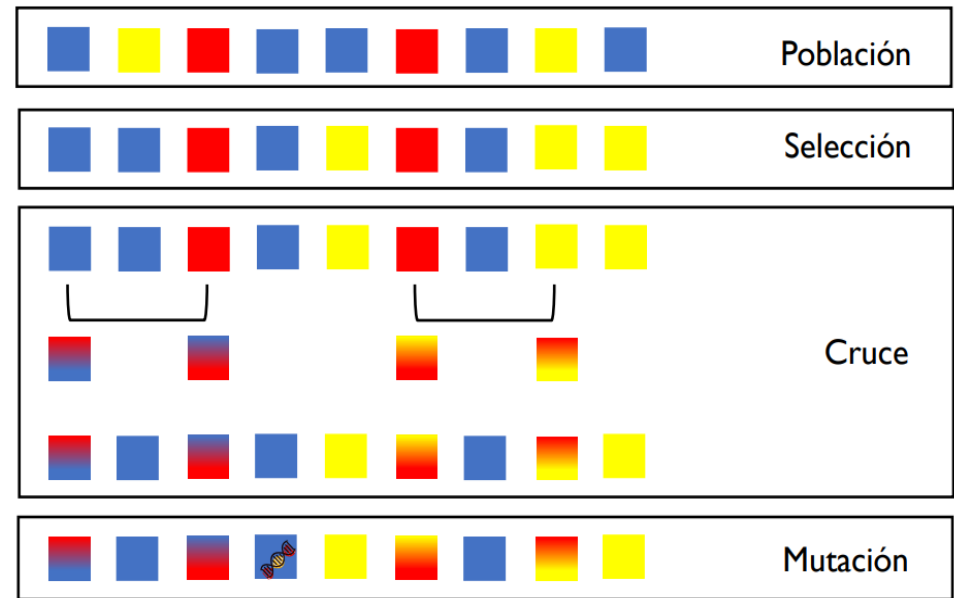
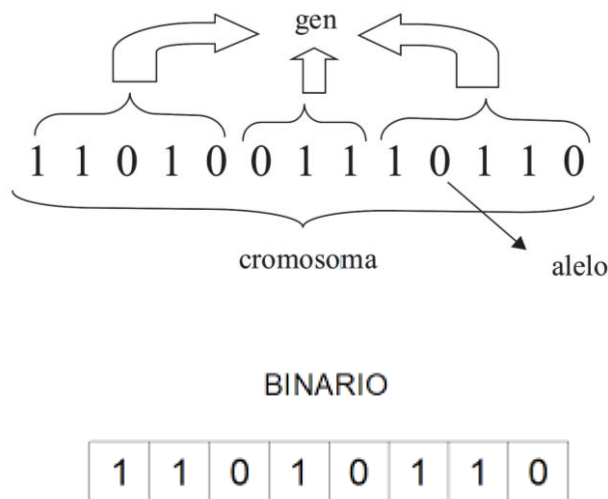


Step 2



Genetic algorithms

It is a heuristic optimization technique based on replicating the mechanism of evolution and adaptation in living organisms.



Genetic algorithm to guide the search

Parameter / Component	Proposed Configuration
Representation	Binary (Chromosomes of length N)
Fitness Function	Accuracy – $\lambda(n/N)$
Parent Selection	Binary Tournament
Crossover Operator	Single-point Crossover
Mutation Operator	Bit-flip Mutation
Elitism	Top 2 individuals pass directly to the next generation
Replacement Mechanism	Generational with Elitism: Offspring replace the previous population while strictly preserving the best individuals to prevent the loss of optimal solutions.

Calibration frame

Parameter	Description	Type	Range min	Range max
n_gen	Number of Generations	Integer (i)	20	100
popSize	Population Size	Integer (i)	20	100
p_cruza	Mating Probability	Real (r)	0.4	0.95
p_mut	Probability of Mutation	Real (r)	0.05	0.2
lambda	Penalty Factor (Parsimony)	Real (r)	0.01	0.2

Parameter	Values
n_gen	94
popSize	64
p_cruza	0.9
p_mut	0.05
lambda	0.01

Comparison Chart

Method	Model Type	Non-linearty	Visualization (Projections)
LDA	Statistical	Linear	Yes
QDA	Statistical	Quadratic	No
SVM (Poly 2)	Kernel-based	Hilbert Space (Degree 2)	No
SVM (Poly 3)	Kernel-based	Hilbert Space (Degree 3)	No
SVM (RBF)	Kernel-based	Hilbert Space (Radial Basis)	No
GMLPDA (Proposed)	stochastic	Non lineal	Yes

Datasets

Dataset	Samples	Variables	Class
iris	150	4	3
penguins	342	4	3
pimadiabetes	768	8	2
wines	178	13	3
Cebolla3D	300	3	3
Cebolla10D	300	10	3
Cebolla20D	300	20	3
Breast Cancer (WDBC)	569	30	2
Ionosphere	351	34	2

Experimental Design

Validation Protocol

- GNLPDA (Stochastic): 30 independent runs X 10-fold cross-validation (N=30 and fixed seed).
- Reference (Deterministic): 1 run $\times$ 10-fold cross-validation (fixed seed).

Statistical Hypothesis

H_0 : Accuracy(GNLPDA) $\leq$ Accuracy(Ref)

H_a : Accuracy(GNLPDA) $>$ Accuracy(Ref)

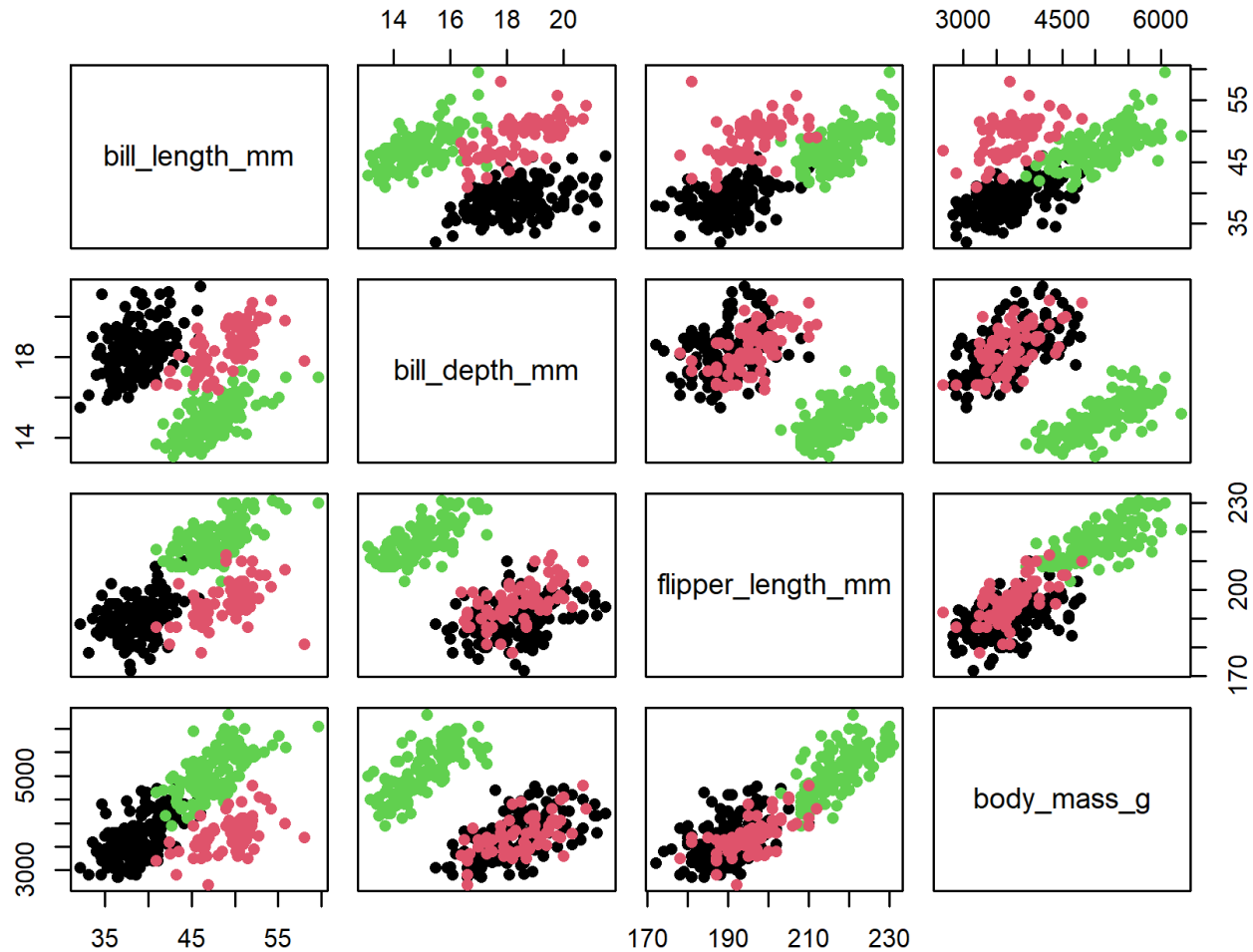
Statistical inference

Test: Wilcoxon Signed-Rank (One-sided).

Confidence level: 95% ($\alpha = 0.05$).

Preliminary results

Case 1: Palmer Penguins

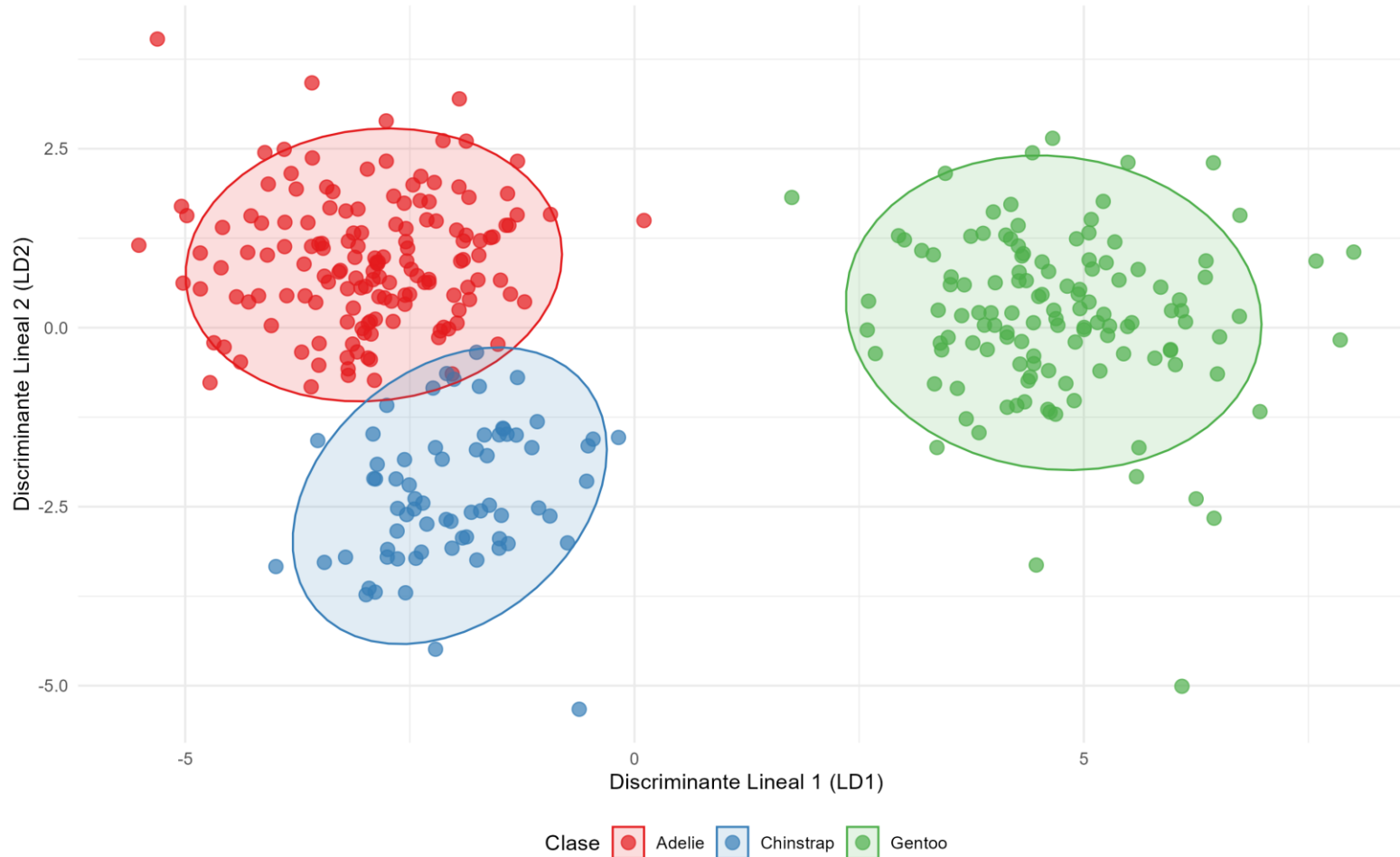


Preliminary results

Case 1: Palmer Penguins

GPN-LDA Projection: 1.2.penguins

Accuracy: 0.991 | Total Vars: 3



Variables
3/4

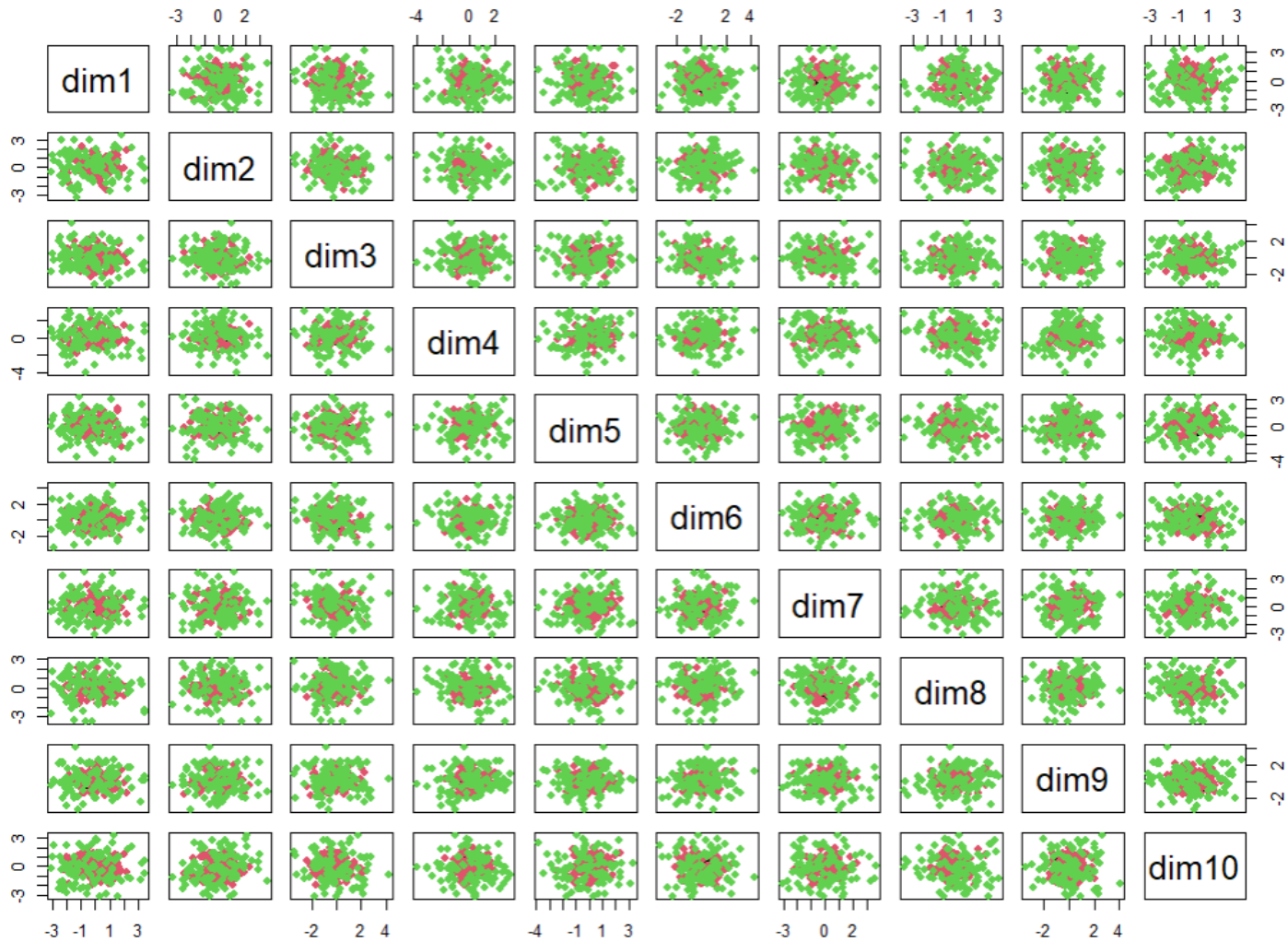
Expanded
terms
3/9

Lin 1: MasaCorporal

*Interact 2: LongitudPico*MasaCorporal, MasaCorporal*AnchoPico*

Preliminary results

Case 2: Cebolla 10D

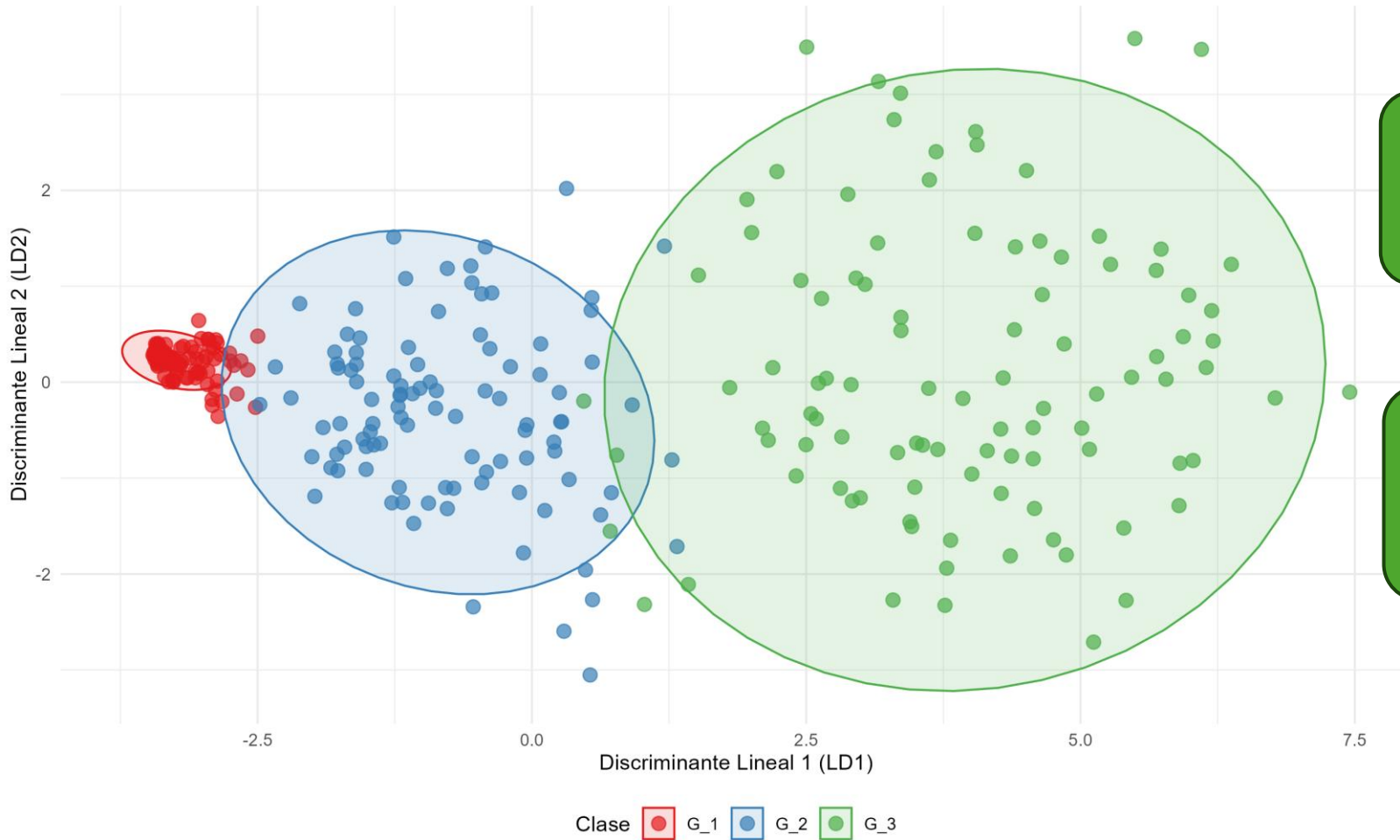


Preliminary results

Case 2: Cebolla 10D

GPN-LDA Projection: 3.2.Cebolla10D

Accuracy: 0.967 | Total Vars: 22



Variables
9/10

Expanded
terms
22/54

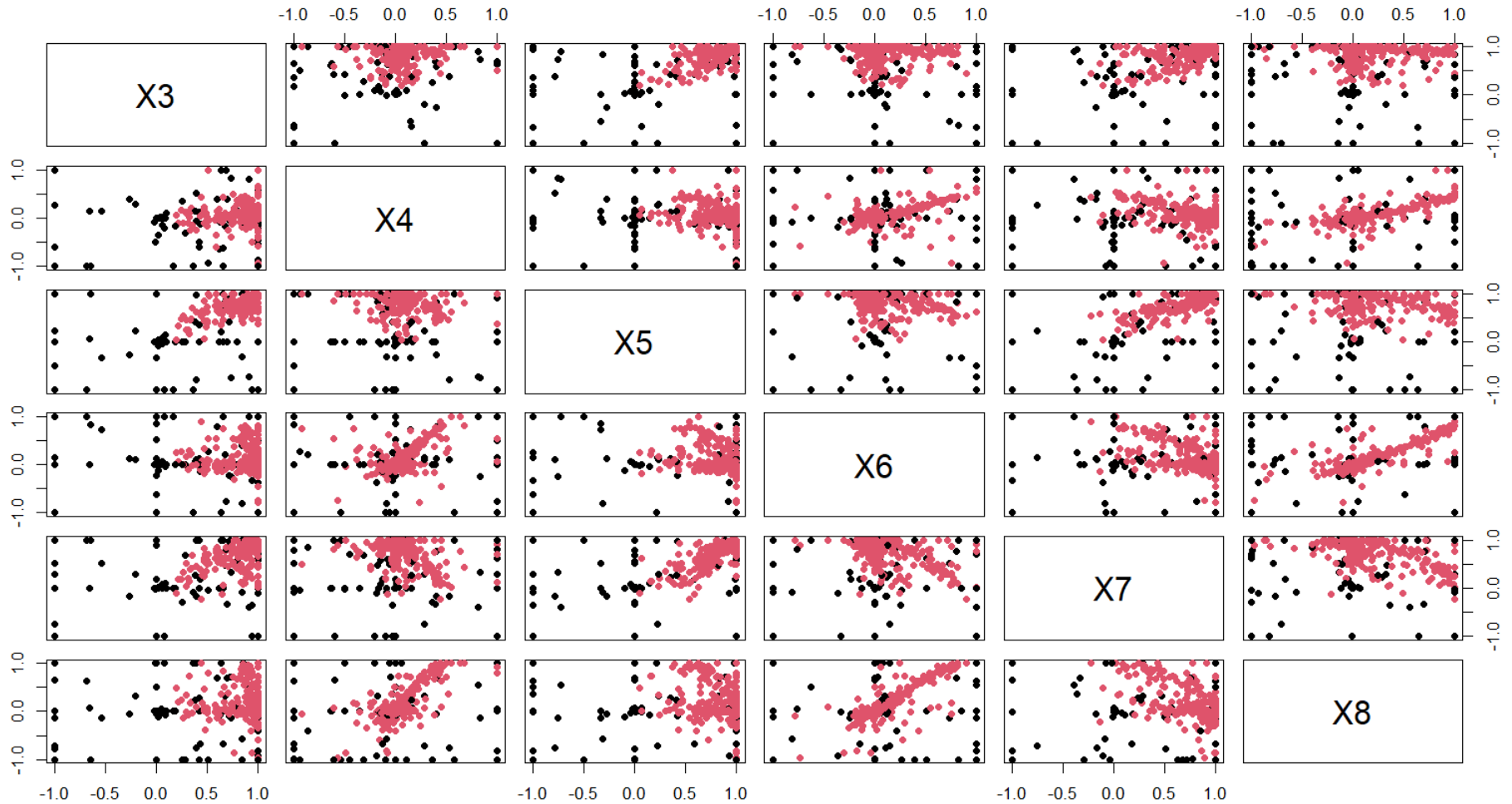
Lin 3: V1, V5, V7

Quad 9: V1², V2², V3², V4², V5², V6², V7², V8², V9²

Interact 10: V3*V8, V5*V4, V5*V6, V6*V1, V6*V2, V7*V2, V7*V8, V9*V4, V9*V6, V9*V8

Preliminary results

Case 3: Ionosphere

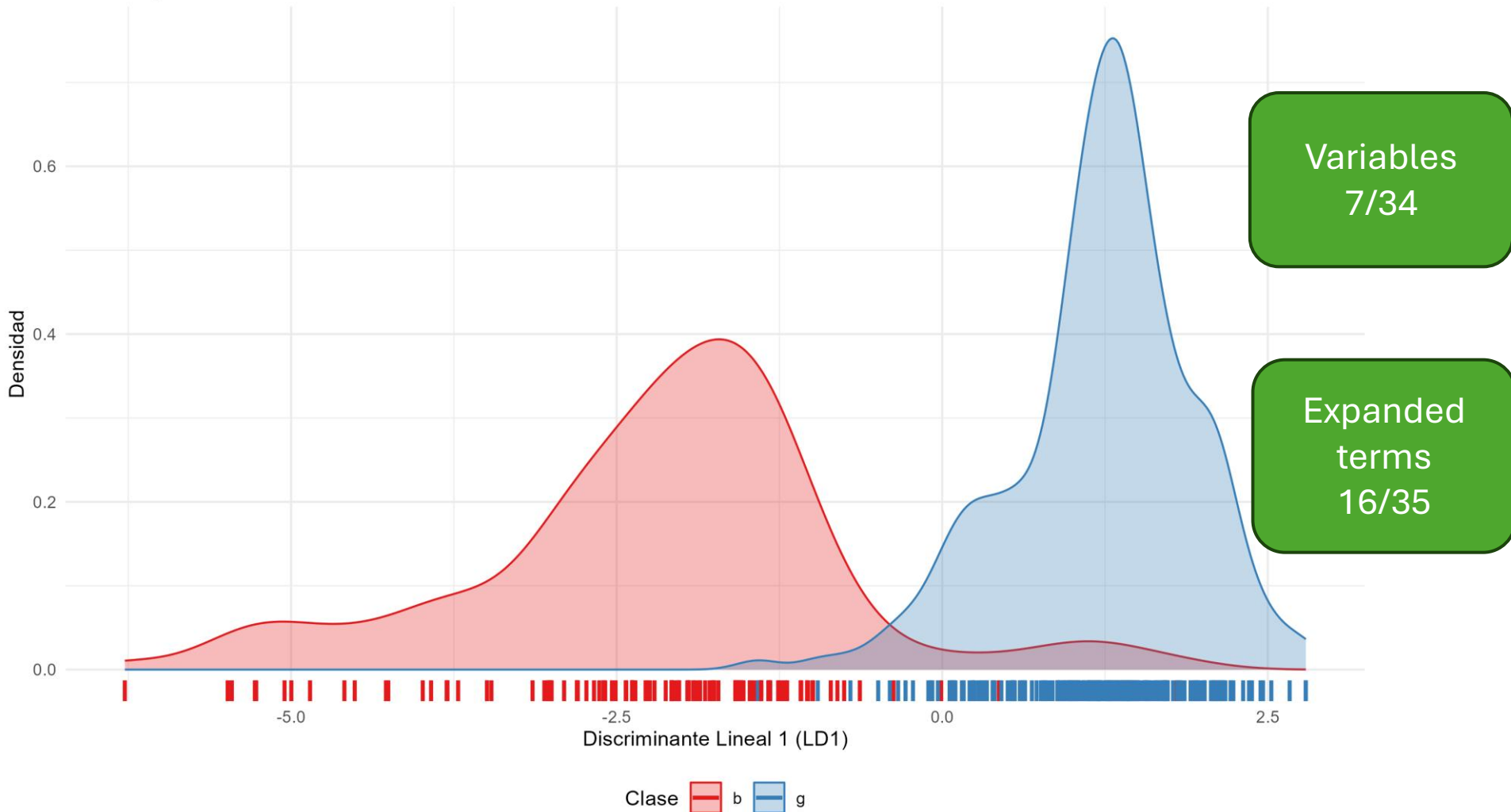


Preliminary results

Case 3: Ionosphere

GPN-LDA Projection: 4.2.Ionosphere

Accuracy: 0.963 | Total Vars: 16



Lin 4: X_{24} , X_5 , X_8 , X_9

Quad 4: X_{15}^2 , X_{16}^2 , X_{27}^2 , X_8^2

Interact 8: $X_{15} \cdot X_{24}$, $X_{15} \cdot X_5$, $X_{15} \cdot X_8$, $X_{15} \cdot X_9$, $X_{27} \cdot X_5$, $X_{27} \cdot X_9$, $X_5 \cdot X_9$, $X_8 \cdot X_9$

Preliminary results

Accuracy comparisons by method

Dataset	LDA	QDA	SVM Poly2	SVM Poly3	SVM GAUSS	GMLPDA
penguins	0.99 ± 0.02	0.99 ± 0.02	0.88 ± 0.06	0.98 ± 0.02	0.98 ± 0.02	0.99 ± 0.00
Cebolla10D	0.44 ± 0.10	0.87 ± 0.05	0.90 ± 0.05	0.51 ± 0.04	0.97 ± 0.03	0.96 ± 0.01
Ionosphere	0.86 ± 0.06	NA $\pm$ NA	0.92 ± 0.04	0.84 ± 0.04	0.95 ± 0.03	0.93 ± 0.01

Preliminary results

Metrics of the proposed algorithm

Dataset	No. Vars	Vars E1	Vars E2	Accuracy	Precision	Recall	F1Score	Time (S)
penguins	4	3.00 ± 0.00	3.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.99 ± 0.00	1.46 ± 1.32
Cebolla10D	10	9.00 ± 0.00	22.33 ± 3.10	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	1892.04 ± 695.36
Ionosphere	34	7.93 ± 1.14	22.90 ± 6.70	0.93 ± 0.01	0.98 ± 0.01	0.84 ± 0.03	0.90 ± 0.02	2282.16 ± 597.35

For the next session

- Present results using repeated k-folds adding statistical tests.

References

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State of the art for Non-linearly

Year	Authors	Method	Focus / Solves	Limitation
2025	Hu et al.	Inter-Spectral Distance	Mass spectra of similar species.	Domain-specific: Limited to spectral data.
2024	Sudibyo et al.	iLDA (Iterative)	Non-Gaussian dimensional reduction.	Unstable: Fails with small sample sizes.
2023	Reza et al.	Integrated RBF Kernel	Complex clinical prediction (PIMA).	Heuristic: Trial-and-error parameter tuning.

State of the art for Dimensional Reduction

Year	Authors	Method	Focus / Solves	Limitation
2025	Sudibyo et al.	DRCT	Ultra-high dimensional reduction.	High Cost: Heavy computational load.
2024	Liu & Guo	Weighted LDA	Imbalanced dataset feature extraction.	Bias Risk: Overfits to minority classes.
2024	Li et al.	Structure Projection	Global/local image feature learning.	Lacks Parsimony: Ignores minimal subsets.

Variable reduction

Dataset	Vars E1 (%)	Vars E2 (%)
penguins	25.00 $\pm$ 0.00	66.67 $\pm$ 0.00
Cebolla10D	10.00 $\pm$ 0.00	58.64 $\pm$ 5.74
Ionosphere	76.67 $\pm$ 3.36	48.05 $\pm$ 7.57